KG COLLEGE OF ARTS AND SCIENCE
Autonomous Institution | Affiliated to Bharathiar University
Accredited with A++ Grade by NAAC
ISO 9001:2015 Certified Institution
KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

INSURANCE POLICY MANAGEMENT AND PREMIUM REMINDER SYSTEM

Submitted by

DHANUSKAASRI P G

25BCS016

Under the Guidance of

Dr. BOOPALAN SUNDARAMOORTHY

Technical Trainer

Technical Department, KG College of Arts and Science

Academic Year: **2025 – 2026**

APRIL 2026

Semester: **II**

DECLARATION

I hereby declare that the project entitled "**INSURANCE POLICY MANAGEMENT AND PREMIUM REMINDER SYSTEM"** submitted in partial fulfillment of the requirements for the course **JAVA PROGRAMMING** is a **record of my original work** carried out under the guidance of **Dr. BOOPALAN SUNDARAMOORTHY**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

DHANUSKAASRI P G

CERTIFICATE

This is to certify that the project entitled **"INSURANCE POLICY MANAGEMENT AND PREMIUM REMINDER SYSTEM"** is a **bona fide work** carried out by **DHANUSKAASRI P G,I - BSC CS 'A' [2025 - 2028], Department Of Computer Science** , **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java Programming**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place:

Date:

Submitted for the Viva-Voce Examination held on _____

Guide:
(Technical Trainer, KGCAS)

_____

Head of the Department
(Technical Head, KGCAS)

_____

TABLE OF CONTENTS

S. No.	Topics	Page. No.
1	**Introduction**	
1.1	Objective	
1.2	Overview	
1.3	Features	
2	**Program Analysis**	
2.1	Problem Definition	
2.2	Project Overview	
2.3	Scopes and Limitations	
3	**System Design**	
3.1	Modules	
4	**Methodology**	
4.1	Problem Analysis	
4.2	System Design	
4.3	Program Development	
4.4	Testing and Validation	
4.5	Result and Documentation	
4.6	Algorithm	
4.7	Flowchart	
5	**Source Code**	
6	**Input and Output Screens**	
7	**Discussion and Conclusion**	
7.1	Discussion	
7.2	Conclusion	
8	**References**	

ABSTRACT

The **Insurance Policy Management and Premium Reminder System** is a software application developed to efficiently manage insurance policies and ensure timely payment of premiums. In the traditional system, managing multiple insurance policies and remembering premium due dates can be difficult and error-prone. This project aims to simplify and automate these tasks.

The system allows users to store and manage policy details such as policy number, policy holder name, type of insurance, premium amount, due date, and contact information. It provides an organized platform for maintaining records and retrieving policy information easily.One of the key features of this system is the premium reminder functionality, which notifies users about upcoming due dates to avoid penalties or policy lapses.

The application is developed using Java programming concepts such as classes, objects, inheritance, and file handling or database connectivity. It ensures data accuracy, reduces manual work, and improves efficiency in managing insurance-related information.

Overall, this system provides a user-friendly and reliable solution for insurance policy tracking and premium payment reminders, making it beneficial for both individuals and organizations.

1. INTRODUCTION

The Insurance Policy Management and Premium Reminder System is a software application designed to simplify the management of insurance policies and ensure timely payment of premiums. In today's fast-paced world, individuals and organizations often hold multiple insurance policies such as life, health, and vehicle insurance. Managing these policies manually can be difficult, time-consuming, and prone to errors.

This system provides a digital platform to store, organize, and manage all insurance-related information in one place. It helps users keep track of policy details such as policy number, type, premium amount, and due dates. One of the major features of this system is the automatic premium reminder, which alerts users before the due date, helping them avoid late fees or policy lapses.

The application is developed using Java programming concepts, making it efficient, reliable, and user-friendly. By automating policy management and reminders, the system reduces manual effort, improves accuracy, and enhances overall efficiency.

1.1 Objective

The main objectives of the project are:

1. To develop an efficient system for managing insurance policies
2. To store and maintain policyholder details in an organized manner
3. To provide a centralized platform for all insurance records
4. To enable easy addition of new insurance policies
5. To allow updating of existing policy information
6. To provide deletion of policy records when necessary
7. To ensure quick and easy retrieval of policy details
8. To implement a premium reminder system for due dates
9. To notify users about upcoming premium payments
10. To reduce the risk of missing premium deadlines
11. To minimize manual errors in policy management
12. To improve data accuracy and consistency
13. To provide a user-friendly interface
14. To ensure secure storage of insurance data
15. To save time and effort compared to manual systems

1.2 Overview

The **Insurance Policy Management and Premium Reminder System** is a software solution designed to simplify the process of managing insurance policies and tracking premium payments. This system provides a centralized platform where users can store, update, and access all policy-related information in an organized manner. It includes essential details such as policy number, policyholder name, insurance type, premium amount, and due dates.

The main purpose of this system is to reduce the difficulties associated with manual record-keeping and to improve efficiency through automation. One of its key features is the premium reminder functionality, which alerts users about upcoming payment deadlines, helping them avoid penalties and policy lapses. The system also allows users to perform operations such as adding new policies, modifying existing records, and deleting outdated information.

Developed using Java programming, the system applies Object-Oriented Programming concepts to ensure reliability, scalability, and maintainability. It offers a user-friendly interface, making it easy for users to interact with the application. Overall, this system provides a secure, efficient, and convenient way to manage insurance policies and ensures timely premium payments.

1.3 Features

Policy Management – Allows users to add, update, and delete insurance policy details easily

Centralized Data Storage – Stores all policy information in one place for easy access

Premium Reminder System – Sends alerts or notifications for upcoming due dates

Search and Retrieval – Enables quick searching of policies using policy number or name

User-Friendly Interface – Simple and easy-to-use design for smooth interaction

Data Accuracy and Validation – Ensures correct data entry and reduces errors

Secure Data Handling – Protects sensitive policyholder information

2. PROGRAM ANALYSIS

Managing insurance policies manually is time-consuming and prone to errors, especially when handling multiple policies. Users often forget premium due dates, leading to penalties or policy lapses. There is a lack of a centralized system to store and retrieve policy information efficiently. Manual record-keeping also increases the risk of data loss and inaccuracies. Therefore, there is a need for an automated system to manage policies and provide timely premium reminders.

2.1 Problem Definition

In the existing manual system, managing insurance policies and tracking premium payments is difficult and inefficient. Users often face challenges in maintaining multiple policy records, remembering due dates, and ensuring timely payments. This may lead to missed premiums, penalties, or policy lapses. Additionally, manual record-keeping increases the chances of data errors and loss of information.

- Managing insurance policies manually is time-consuming
- Users may forget premium due dates
- Lack of a centralized system
- Increased chances of errors and data loss
- Difficulty in retrieving policy details
- Inefficient tracking of multiple policies
- Need for an automated solution

2.2 Project Overview

The **Insurance Policy Management and Premium Reminder System** is a Java-based application designed to manage insurance policies efficiently. It provides a centralized platform to store and access policy details such as policy number, policyholder name, premium amount, and due dates.

The system helps users manage multiple policies easily and includes a premium reminder feature to notify users about upcoming payment deadlines, preventing penalties and policy lapses. It also allows adding, updating, deleting, and searching policy records.

Overall, the project reduces manual work, improves accuracy, and provides a simple and efficient solution for managing insurance policies.

2.3 Scopes and Limitations

Scope

- The system can be used by **individuals and small organizations** to manage insurance policies
- It provides a platform to **store, update, and retrieve policy information**
- It supports management of **multiple types of insurance** (life, health, vehicle, etc.)
- The system can be enhanced to include **SMS or email reminder notifications**
- It can be extended with **online payment integration** for premium payments
- The application can be upgraded to support **cloud-based storage**
- It can be integrated with **mobile applications** for easy access
- The system can be expanded to include **advanced reporting and analytics**
- It has the potential to support **multi-user access with authentication**
- The project can be further developed into a **full-scale enterprise application**

Limitation

- It may not include real-time SMS or email notifications (basic reminders only)
- Requires manual data entry, which may still lead to minor errors
- Limited security features compared to advanced systems
- May not support online payment integration
- The system might be dependent on a single device or local storage
- Limited scalability for large organizations
- Does not include advanced analytics or reporting features
- Requires basic technical knowledge to operate
- No automatic data backup unless implemented separately

3. SYSTEM DESIGN

The Insurance Policy Management and Premium Reminder System is designed using a modular approach to ensure simplicity, efficiency, and easy maintenance. The system consists of a user interface, processing logic, and data storage**.**

- **User Interface (UI):**
  Provides interaction between the user and the system. Users can enter, update, and view policy details through a simple interface**.**
- **Application Logic:**
  Handles all operations such as adding, updating, deleting, and searching policy records. It also manages the reminder functionality.
- **Database / Storage:**
  Stores all policy-related data such as policy number, policyholder name, premium amount, and due dates.
- **Reminder Mechanism:**
  Checks due dates and alerts users about upcoming premium payments.

3.1 Modules

1. **User Module**
   - Handles user interaction with the system
   - Allows users to access and manage policy details
2. **Policy Management Module**
   - Add new insurance policies
   - Update existing policy details
   - Delete policy records
3. **Premium Reminder Module**
   - Tracks premium due dates
   - Sends reminders for upcoming payments
4. **Search Module**
   - Allows searching policies by policy number or name
   - Provides quick retrieval of data
5. **Data Storage Module**
   - Stores all policy-related information
   - Ensures data is saved and retrieved efficiently
6. **Validation Module**
   - Checks correctness of input data
   - Prevents invalid or incomplete entries

4. METHODOLOGY

The **Insurance Policy Management and Premium Reminder System** is developed using a structured and step-by-step approach to ensure efficiency and reliability. Initially, the requirements of the system are identified, focusing on managing policy details and tracking premium due dates. Based on these requirements, the system is designed with different modules such as policy management, reminder system, and data storage.

The development process includes the following stages:

• Problem Analysis
• System Design
• Program Development
• Testing and Validation
• Result and Documentation

4.1. Problem Analysis

In this stage, the problems of the existing manual system are identified.
The main issues include:

- Difficulty in managing multiple insurance policies
- Forgetting premium due dates
- Lack of a centralized system
- Risk of data loss and errors

4.2. System Design

In this stage, the overall structure of the system is designed before development.
The system is divided into modules such as:

- Policy Management Module
- Premium Reminder Module
- Search Module
- Data Storage Module
- User Interface Module

4.3. Program Development

In this stage, the system is developed using Java programming.
The following operations are implemented:

- Adding new policy details
- Updating existing records
- Deleting policy information
- Searching policy data
- Implementing reminder functionality

4.4. Testing and Validation

In this stage, the system is tested to ensure correct functioning.
The following checks are performed:

- Verification of input and output data
- Testing reminder alerts
- Identifying and fixing errors
- Ensuring all modules work properly

4.5. Result and Documentation

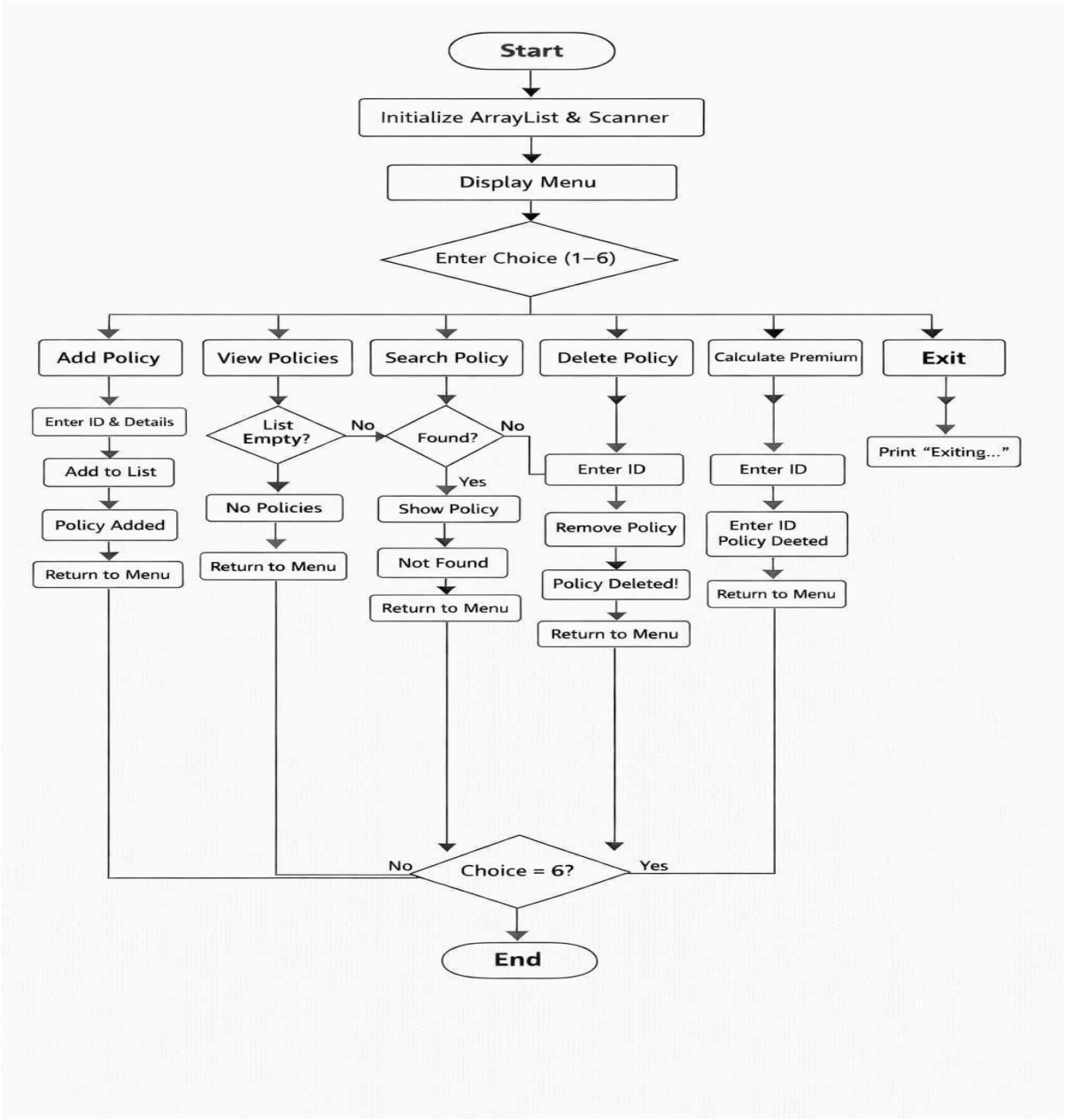
In this stage, the final output of the system is verified.
The following activities are carried out:

- Displaying correct policy details
- Ensuring proper working of reminders
- Documenting system design and implementation
- Recording outputs and results for reference

4.6. Algorithm

1. Start
2. Display menu options:
    - Add Policy
    - View Policies
    - Search Policy
    - Delete Policy
    - Exit
3. Enter user choice
4. If choice = Add Policy
    - Enter policy details (ID, Name, Type, Premium, Due Date)
    - Store the details
5. If choice = View Policies
    - Display all stored policies
6. If choice = Search Policy
    - Enter policy ID
    - Display matching policy details
7. If choice = Delete Policy
    - Enter policy ID
    - Remove the policy
8. If choice = Exit
    - Stop the program
9. Else
    - Display invalid choice
10. Repeat until user exits
11. Stop

4.7. Flowchart



5. SOURCE CODE

```
import java.util.*;
class Policy {
    int id;
    String name;
    String type;
    double premium;

    Policy(int id, String name, String type, double premium) {
        this.id = id;
        this.name = name;
        this.type = type;
        this.premium = premium;
    }
       double calculateYearlyPremium() {
        Return premium * 12;// monthly -> yearly
    }
}

public class InsuranceSystem {

    static ArrayList<Policy> policies = new ArrayList<>();
    static Scanner sc = new Scanner(System.in);

    public static void main(String[] args) {

        int choice;
```

```
do {
    System.out.println("\n===== Insurance Policy Management =====");
    System.out.println("1. Add Policy");
    System.out.println("2. View Policies");
    System.out.println("3. Search Policy");
    System.out.println("4. Delete Policy");
    System.out.println("5. Calculate Premium (Billing)");
    System.out.println("6. Exit");
    System.out.print("Enter your choice: ");
    choice = sc.nextInt();

    switch (choice) {
        case 1:
            addPolicy();
            break;
        case 2:
            viewPolicies();
            break;
        case 3:
            searchPolicy();
            break;
        case 4:
            deletePolicy();
            break;
        case 5:
            calculatePremium();
            break;
        case 6:
            System.out.println("Exiting...");
```

```
                break;
            default:
                System.out.println("Invalid choice!");
        }

    } while (choice != 6);
}

static void addPolicy() {
    System.out.print("Enter Policy ID: ");
    int id = sc.nextInt();
    sc.nextLine();

    System.out.print("Enter Name: ");
    String name = sc.nextLine();

    System.out.print("Enter Policy Type: ");
    String type = sc.nextLine();

    System.out.print("Enter Monthly Premium: ");
    double premium = sc.nextDouble();

    policies.add(new Policy(id, name, type, premium));
    System.out.println("Policy Added Successfully!");
}

static void viewPolicies() {
    for (Policy p : policies) {
```

```
            System.out.println(p.id + " | " + p.name + " | " + p.type + " | Monthly: " +
p.premium);
        }
    }

    static void searchPolicy() {
        System.out.print("Enter Policy ID to search: ");
        int id = sc.nextInt();

        for (Policy p : policies) {
            if (p.id == id) {
                System.out.println("Found: " + p.name + " | " + p.type);
                return;
            }
        }
        System.out.println("Policy Not Found!");
    }

    static void deletePolicy() {
        System.out.print("Enter Policy ID to delete: ");
        int id = sc.nextInt();

        policies.removeIf(p -> p.id == id);
        System.out.println("Policy Deleted!");
    }

    static void calculatePremium() {
        System.out.print("Enter Policy ID: ");
        int id = sc.nextInt();
```

```
        for (Policy p : policies) {
            if (p.id == id) {
                double yearly = p.calculateYearlyPremium();
                System.out.println("Yearly Premium for " + p.name + " = " + yearly);
                return;
            }
        }
        System.out.println("Policy Not Found!");
    }
}
```

INPUT AND OUTPUT SCREEN

```
===== Insurance Policy Management =====
1. Add Policy
2. View Policies
3. Search Policy
4. Delete Policy
5. Calculate Premium (Billing)
6. Exit
Enter your choice: 1
Enter Policy ID: 101
Enter Name: dhanuskaasri
Enter Policy Type: Life Insurence
Enter Monthly Premium: 5000
Policy Added Successfully!

===== Insurance Policy Management =====
1. Add Policy
2. View Policies
3. Search Policy
4. Delete Policy
5. Calculate Premium (Billing)
6. Exit
Enter your choice: 1
Enter Policy ID: 102
Enter Name: Achu
Enter Policy Type: car insurence
Enter Monthly Premium: 5000
Policy Added Successfully!

===== Insurance Policy Management =====
1. Add Policy
2. View Policies
3. Search Policy
4. Delete Policy
5. Calculate Premium (Billing)
6. Exit
Enter your choice: 2
101 | dhanuskaasri | Life Insurence | Monthly: 5000.0
102 | Achu | car insurence | Monthly: 5000.0
```

```
===== Insurance Policy Management =====
1. Add Policy
2. View Policies
3. Search Policy
4. Delete Policy
5. Calculate Premium (Billing)
6. Exit
Enter your choice: 3
Enter Policy ID to search: 101
Found: dhanuskaasri | Life Insurence

===== Insurance Policy Management =====
1. Add Policy
2. View Policies
3. Search Policy
4. Delete Policy
5. Calculate Premium (Billing)
6. Exit
Enter your choice: 4
Enter Policy ID to delete: 102
Policy Deleted!

===== Insurance Policy Management =====
1. Add Policy
2. View Policies
3. Search Policy
4. Delete Policy
5. Calculate Premium (Billing)
6. Exit
Enter your choice: 5
Enter Policy ID: 101
Yearly Premium for dhanuskaasri = 60000.0
```

```
===== Insurance Policy Management =====
1. Add Policy
2. View Policies
3. Search Policy
4. Delete Policy
5. Calculate Premium (Billing)
6. Exit
Enter your choice: 6
Exiting...

C:\Users\KGCAS\Documents\java prgs\program>
```

6. DISCUSSION & CONCLUSION

7.1 Discussion

The development of this project highlights the importance of automation in managing insurance-related tasks. The system simplifies complex processes such as maintaining policy records and tracking premium payments. It demonstrates the practical use of programming concepts like classes, objects, loops, and data structures.

Although the system performs basic operations effectively, it has some limitations such as lack of real-time reminders and database integration. These limitations can be improved in future by adding features like SMS/email notifications, online payment options, and cloud storage.

Overall, the project provides a strong foundation for further development and can be enhanced into a full-scale application for real-world use.

7.2 Conclusion

The **Insurance Policy Management and Premium Reminder System** successfully provides an efficient solution for managing insurance policies in a simple and organized manner. The system allows users to add, view, search, and delete policy details easily. It also includes a premium calculation feature that helps users understand their yearly payment.

By using Java and Object-Oriented Programming concepts, the system ensures better data handling, accuracy, and reliability. The project reduces manual effort, minimizes errors, and improves efficiency in managing multiple policies. Overall, the system meets its objectives and provides a user-friendly and effective solution for insurance management.

REFERENCES

The following resources were used during the development of this project:

Books

Herbert Schildt, *Java: The Complete Reference*, McGraw-Hill Education

E. Balagurusamy, *Programming with Java*, McGraw-Hill Education

Kathy Sierra & Bert Bates, *Head First Java*, O'Reilly Publications

Cay S. Horstmann, *Core Java Volume I – Fundamentals*, Prentice Hall

Y. Daniel Liang, *Introduction to Java Programming*, Pearson

Websites

Oracle Java Documentation
https://docs.oracle.com/javase/tutorial/

W3Schools Java Tutorial
https://www.w3schools.com/java/

GeeksforGeeks – Java Programming
https://www.geeksforgeeks.org/java/

TutorialsPoint – Java
https://www.tutorialspoint.com/java/

Javatpoint – Java Tutorial
https://www.javatpoint.com/java-tutorial